

WEB DEVELOPMENT

SE102.3

CHAPTER 05

PHP – Part A

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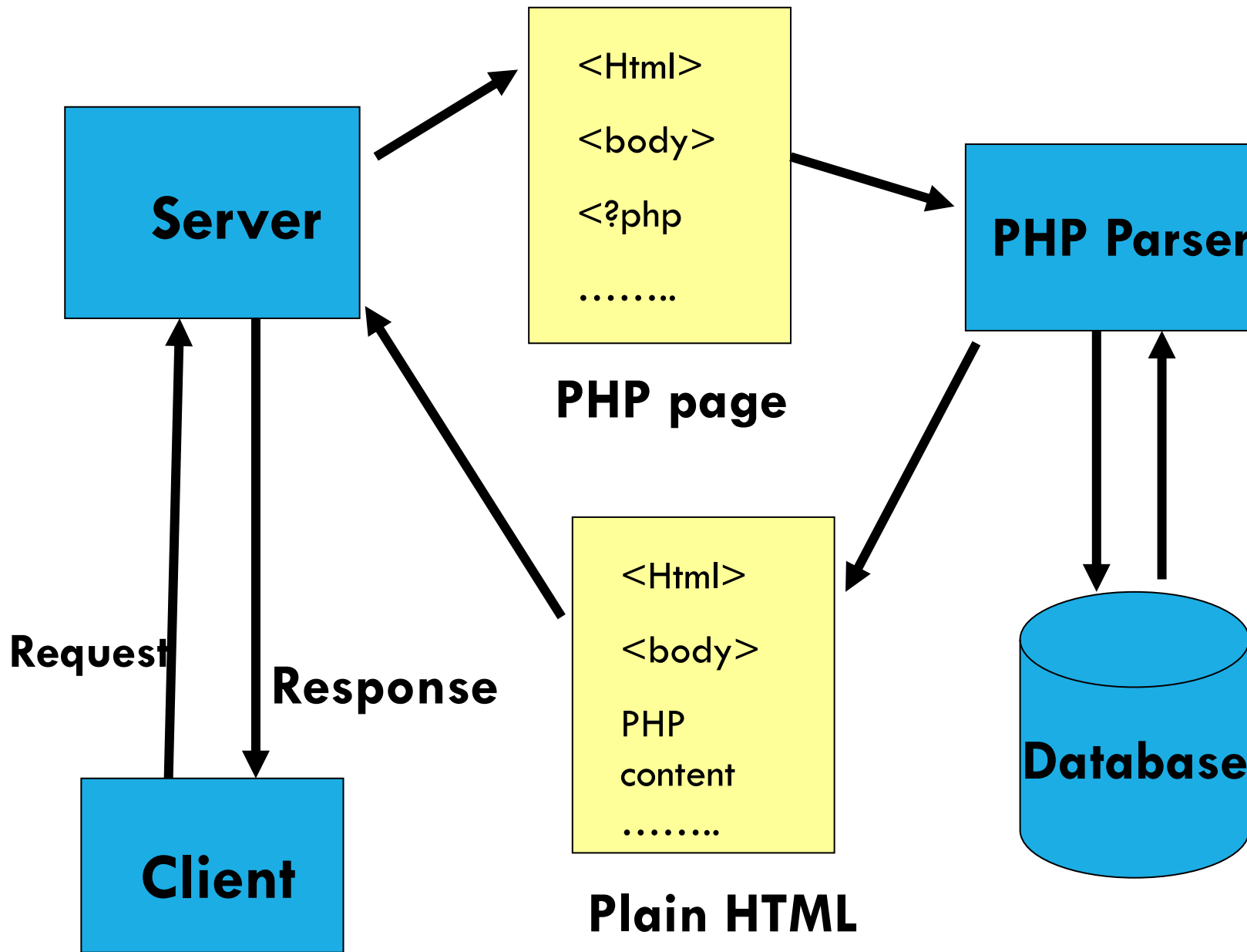
What is PHP?

- PHP is a popular scripting language used for creating dynamic web pages and web applications.
- The term PHP is an acronym of Hypertext Preprocessor.
- It is an open-source, interpreted, object-oriented server-side scripting language.
- PHP scripts can be easily embedded into HTML.
- Files are saved with “.php” extension. Every PHP line terminates with a semicolon.

Features of PHP

- Open-Source and Free: PHP is firstly open source which means anyone can use PHP code without any licensing. Along with this one can run PHP on any operating system like Windows, macOS, Linux, Unix and more.
- PHP Server-Side Scripting: PHP code executes on the server before sending HTML content to the user's browser, allowing for the dynamic generation of web pages and handling user interactions.

- Database connectivity: PHP integrates seamlessly with various databases like MySQL, PostgreSQL, and Oracle, facilitating data storage and retrieval for web applications.
- Object-oriented programming (OOP): PHP supports OOP concepts like classes, objects, inheritance, and polymorphism, enabling better code organization and modularity.
- Built-in functions: PHP comes with a rich set of built-in functions for various tasks such as string manipulation, date and time handling, file handling, and more, reducing the need for external libraries.
- Session management: PHP allows for user session management, enabling personalized experiences and storing user data across multiple page visits.
- Security features: While security considerations are essential for any development language, PHP offers several built-in security features and best practices to help mitigate vulnerabilities.



When a web browser requests a PHP page from the server, the server calls up the PHP parser. The parser executes all the PHP script instructions on the page and returns the file to the browser as plain HTML. During this process the PHP parser may also retrieve information from a database.

In order to develop and run PHP web pages three main components are needed:

Web Server

PHP will work with many of the Web Servers such as IIS. But is most often used with the freely available Apache server.

Database

PHP supports many databases such as MYSQL, Oracle, Informix, Sybase etc.

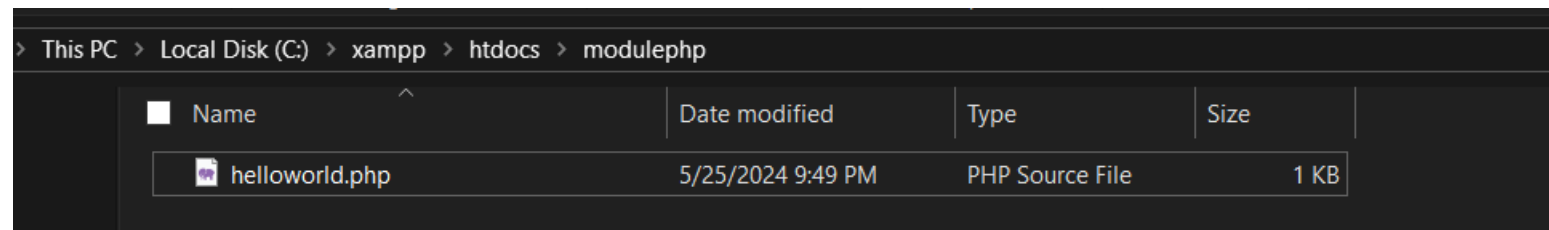
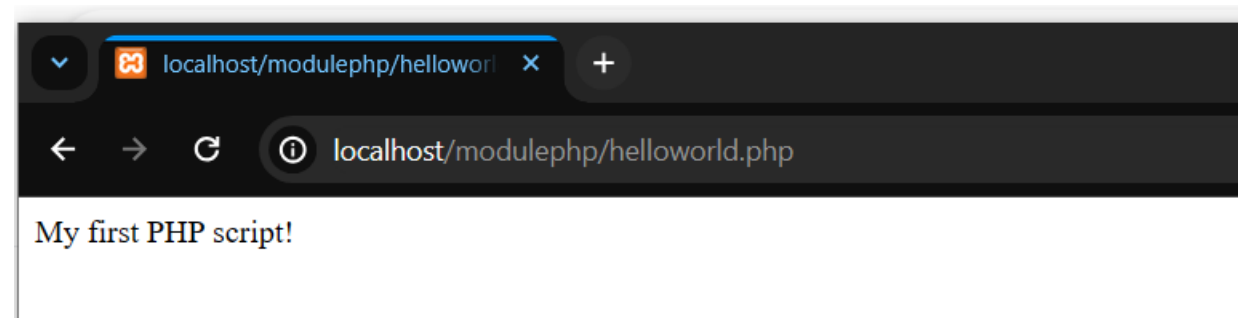
PHP Parser

In order to process PHP script instructions to generate HTML a parser is required.

PHP Syntax

```
<!DOCTYPE html>
<html>
<body>
    <?php
    echo "My first PHP script!";
    ?>
</body>
</html>
```

PHP Execution



Comments

- Comments can be added inside a PHP code block by using `//` and `#` between single line comments.
- `/*` and `*/` between multi-line comments.

Variables

- All variables in PHP are denoted with a leading dollar sign (\$).
- A variable can have a short name (like \$x and \$y) or a more descriptive name (\$age, \$carname, \$total_volume).
- A variable starts with the \$ sign, followed by the name of the variable
- A variable name must start with a letter or the underscore character
- A variable name cannot start with a number
- A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
- Variable names are case-sensitive (\$age and \$AGE are two different variables)

Variable Types

PHP has no command for declaring a variable, and the data type depends on the value of the variable.

PHP supports the following data types:

- String
- Integer
- Float (floating point numbers - also called double)
- Boolean
- Array
- Object
- NULL
- Resource

```
$x = 5;           // $x is an integer  
$y = "John";     // $y is a string  
$z = 3.14;       // $z is a float value  
$r = true;       // $r is a boolean value
```

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
  <?php
```

```
    $name = "Janish De Silva";
```

```
    echo "My name is $name!";
```

```
  ?>
```

```
</body>
```

```
</html>
```

Example

Write down a sentence to display your name, university and age. Declare three variables.

Output –

My name is **Janish De Silva**. I'm an undergraduate of **NSBM Green University**.
I'm **16** years old.

```
<!DOCTYPE html>
<html>
<body>
    <?php
    $name = "Janish De Silva";
    $university = "NSBM Green University";
    $age = "16";

    echo "My name is $name. I'm an undergraduate of $university. I'm $age years old.
    ";
    ?>

</body>
</html>
```

Example – Variable Declaration

You are comparing prices of one item at two different shops to find the best deals.

Declare variables for the shop names, the one items included, the prices of each item at both shops.

Variables:

Shop 1 Name: `$shop1_name`

Shop 2 Name: `$shop2_name`

Item : `$item`

Price of Item at Shop 1: `$price_item1`

Price of Item at Shop 2: `$price_item2`

```
<?php
// Shop names
$shop1_name = "Shop A";
$shop2_name = "Shop B";

// Item
$item = "Laptop";

// Prices of the item at both shops
$price_item1 = 1200; // Price at Shop 1
$price_item2 = 1150; // Price at Shop 2

// Displaying the information
echo "Shop 1: $shop1_name\n";
echo "Shop 2: $shop2_name\n";
echo "Item: $item\n";
echo "Price of $item at $shop1_name: $$price_item1\n";
echo "Price of $item at $shop2_name: $$price_item2\n";
?>
```

PHP Arithmetic Operators

Operator	Name	Example	Result
+	Addition	$\$x + \y	Sum of $\$x$ and $\$y$
-	Subtraction	$\$x - \y	Difference of $\$x$ and $\$y$
*	Multiplication	$\$x * \y	Product of $\$x$ and $\$y$
/	Division	$\$x / \y	Quotient of $\$x$ and $\$y$
%	Modulus	$\$x \% \y	Remainder of $\$x$ divided by $\$y$
**	Exponentiation	$\$x ** \y	Result of raising $\$x$ to the $\$y$ 'th power

PHP if Statements

In PHP we have the following conditional statements:

if statement - executes some code if one condition is true

if...else statement - executes some code if a condition is true and another code if that condition is false

if...elseif...else statement - executes different codes for more than two conditions

switch statement - selects one of many blocks of code to be executed

IF Condition

Write a PHP script that checks the temperature and displays the message "It's a sunny day!" if the temperature is greater than 30 degrees Celsius.

Declare a variable for the temperature as 35.

Use an if statement to determine if the message should be displayed.

```
<?php
```

```
// Temperature in degrees Celsius
```

```
$temperature = 32;
```

```
// Check if the temperature is greater than 30 degrees
```

```
if ($temperature > 30) {
```

```
    echo "It's a sunny day!";
```

```
}
```

```
?>
```

IF.. Else Condition

Write a PHP script that checks the age of a person and displays a message based on their age.

If the age is 18 or older, display "You are eligible to vote!". If the age is less than 18, display "You are not eligible to vote.".

Assign the current age into 20 years.

```
<?php

// Age of the person
$age = 20;

// Check the age and display the appropriate message

if ($age >= 18) {
    echo "You are eligible to vote!";
} else {
    echo "You are not eligible to vote.";
}
?>
```

IF...ELSEIF...ELSE statement

Write a PHP script that checks a student's grade and displays a message based on the grade.

If the grade is 90 or above, display "Excellent!".

If the grade is between 75 and 89, display "Good job!".

If the grade is between 50 and 74, display "You passed."

If the grade is below 50, display "You failed."

Assign the student's grade in to 72.

```
<?php
```

```
// Grade of the student
```

```
$grade = 72;
```

```
if ($grade >= 90) {  
    echo "Excellent!";  
} elseif ($grade >= 75) {  
    echo "Good job!";  
} elseif ($grade >= 50) {  
    echo "You passed.";  
} else {  
    echo "You failed.";  
}  
?>
```

Example

Write a PHP script that determines the category of a person based on their age.

Use the following categories:

If the age is 0-12, display "You are a child".

If the age is 13-19, display "You are a teenager.

If the age is 20-64, display "You are an adult."

If the age is 65 or above, display "You are a senior."

Assign the person age as 15 years.


```
<?php
```

```
// Age of the person
```

```
$age = 15;
```

```
if ($age >= 0 && $age <= 12) {  
    echo "You are a child.";  
} elseif ($age >= 13 && $age <= 19) {  
    echo "You are a teenager.";  
} elseif ($age >= 20 && $age <= 64) {  
    echo "You are an adult.";  
} else {  
    echo "You are a senior.";  
}  
?>
```

Switch Statement – Example

Imagine you're creating a PHP script for a quiz application where users select their preferred programming language.

Write a PHP switch statement that outputs a message based on the user's choice of programming language (`$language`).

Handle at least three programming languages in your cases (e.g., PHP, Python, JavaScript), with a default case for unrecognized choices.

```
<?php
```

```
$language = "PHP"; // Assume this value is obtained from user input
```

```
switch ($language) {  
    case "PHP":  
        echo "You have chosen PHP";  
        break;  
    case "Python":  
        echo "You have chosen Python";  
        break;  
    case "JavaScript":  
        echo "You have chosen JavaScript";  
        break;  
    default:  
        echo "You have chosen a programming language that is not supported in this quiz.";  
}  
  
?>
```

PHP Arrays

An array is a special variable that can hold many values under a single name, and you can access the values by referring to an index number or name.

An array stores multiple values in one single variable:

```
$cars = array("Volvo", "BMW", "Toyota");
```

Example

You have a PHP script embedded within an HTML document that defines an array `$cars` containing several car brands.

Explain what the `var_dump($cars);` function does in this context and what output you would expect to see when you view the HTML page in a browser.

```
<?php
$cars = array("Volvo", "BMW", "Toyota");
var_dump($cars);
?>
```

```
array(3) {
    [0]=>
    string(5) "Volvo"
    [1]=>
    string(3) "BMW"
    [2]=>
    string(6) "Toyota"
}
```

PHP Loops

In PHP, we have the following loop types:

- **while - loops** through a block of code as long as the specified condition is true
- **do...while** - loops through a block of code once, and then repeats the loop as long as the specified condition is true
- **for** - loops through a block of code a specified number of times
- **foreach** - loops through a block of code for each element in an array.

PHP Functions

Besides the built-in PHP functions, it is possible to create your own functions.

A function is a block of statements that can be used repeatedly in a program.

A function will not execute automatically when a page loads.

A function will be executed by a call to the function.

```
<?php
```

```
function myMessage() {  
    echo "Hello world!";  
}  
myMessage();
```

```
?>
```

PHP Function Arguments

```
<?php
function friendsName($fname)
{
    echo "$fname";
}

friendsName("Jane \n");
friendsName("Hegeni \n");
friendsName("Stoli \n");
friendsName("Jim \n");
friendsName("Borge \n");

?>
```


PHP Function Arguments

The friendName function takes two parameters: \$fname and \$year. It echoes a string that includes the friend's name and their birth year.

```
<?php
function friendName($fname, $year) {
    echo "$fname is my friend. Born in $year \n";
}

friendName("Hege", "1975");
friendName("Stale", "1978");

?>
```

PHP Default Argument Value

```
<?php

function setweight($minweight = 50)
{
    echo "The weight is : $minweight \n";
}

setweight(350);
setweight();

?>
```

PHP Functions - Returning values

```
<?php
function sum($x, $y) {
    $z = $x + $y;
    return $z;
}

echo "5 + 10 =" . sum(5,10). "\n ";
echo sum(2,4) ;

?>
```

Exercise

In a shop, there are 5 apples. The price of an apple is 100 rupees.

Write a PHP script using functions to calculate and display the total value of the apples in the shop.

```
<?php
// Function to calculate the total value of apples
function calculateTotalValue($num_apples, $price_per_apple) {
    // Calculate the total value inside the function
    $total_value = $num_apples * $price_per_apple;
    // Return the calculated total value
    return $total_value;
}

// Number of apples
$num_apples = 5;

// Price per apple
$price_per_apple = 100;

// Call the function to calculate the total value
$total_value = calculateTotalValue($num_apples, $price_per_apple);

// Display the total value
echo "The total value of apples in the shop is: $total_value rupees.";

?>
```

Exercise

In a shop, there are 8 mangoes. The price of a mango is 75 rupees.

Write a PHP script using functions to calculate and display the difference in value compared to another shop where there are 5 mangoes priced at 60 rupees each.

```
<?php
```

```
// Function to calculate the difference in value
```

```
function calculateDifference ($num_mangoes_shop1, $price_per_mango_shop1,  
$num_mangoes_shop2, $price_per_mango_shop2)
```

```
{
```

```
    $total_value_shop1 = $num_mangoes_shop1 * $price_per_mango_shop1;
```

```
    $total_value_shop2 = $num_mangoes_shop2 * $price_per_mango_shop2;
```

```
    // Calculate the difference in value
```

```
    $difference = $total_value_shop1 - $total_value_shop2;
```

```
    return $difference;
```

```
}
```

```
// Shop 1
$num_mangoes_shop1 = 8;
$price_per_mango_shop1 = 75;

// Shop 2
$num_mangoes_shop2 = 5;
$price_per_mango_shop2 = 60;

// Call the function to calculate the difference
$difference_value = calculateDifference($num_mangoes_shop1, $price_per_mango_shop1,
$num_mangoes_shop2, $price_per_mango_shop2);

// Display the difference in value
echo "The difference in value between the two shops is: $difference_value rupees.";

?>
```